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Question	Answer	Marks
1	**Defining the problem**	
	m is the independent variable and θ is the dependent variable **or** vary *m* and measure θ **or** θ is the independent variable and *m* is the dependent variable **or** vary θ and measure *m*	**1**
	keep position of load constant or *L* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • load shown touching at P • load labelled and at least one other label	**1**
	use a protractor to measure θ	**1**
	if *m* is the independent variable: (slowly) change the angle until the block (just) topples **or** if θ is the independent variable: (slowly) change mass until block (just) topples	**1**
	use a balance to measure *m*	**1**

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1	**Method of analysis**	
	plot a graph of cos θ against 1 / m (**or** reverse axes 1 / m against cos θ) (**not** 1 / cos θ against m or m against 1 / cos θ)	**1**
	relationship valid if a straight line (**not** 'through the origin' unless the choice of axes indicates the intercept is zero)	**1**
	$\alpha = \dfrac{2L \times \text{gradient}}{V \times w}$ **or** $\alpha = \dfrac{\text{gradient}}{V \times y\text{-intercept}}$ for reverse axes: $\alpha = \dfrac{2L}{V \times w \times \text{gradient}}$ **or** $\alpha = -\dfrac{1}{V \times y\text{-intercept}}$	**1**